

- 3 (a) The principle of superposition states that when two waves of the same type meet at a point in space, the resultant displacement at that point is equal to the vector sum of the displacements that each wave would produce at that point.

Note: A common mistake is using amplitude, rather than displacement.

(b) (i) Using $x = \frac{\lambda D}{a}$, we have

$$x = \frac{590 \times 10^{-9} \times 2.3}{1.2 \times 10^{-3}}$$

$$x \approx 1.13 \times 10^{-3} \text{ m}$$

(ii) Since $I \propto A^2$, for $I' = \frac{I}{3}$, the amplitude of the waves from slit B is $\frac{1}{\sqrt{3}} A$.

At point P, the resultant amplitude of the wave interfering destructively is

$$A_{\text{resultant}} = A - \frac{1}{\sqrt{3}} A$$

$$A_{\text{resultant}} \approx 0.423A$$

Hence, the resultant intensity is $I_{\text{resultant}} \propto (0.423A)^2 = 0.179I$

4. (a) When electrons interact with the metal target atoms, they decelerate and can lose